ITCS254 | Lab 6: Processes | September 7, 2026

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Paste captured screenshots of the results for each question.

1. List all process details, including PPID

```
parallels@ubuntu-gnu-linux-24-04-3:~$ ps lf
F   UID     PID    PPID PRI  NI    VSZ   RSS WCHAN  STAT TTY        TIME COMMAND
0  1000    2884    2877  20   0  12736  6100 do_wai Ss   pts/0      0:00 bash
0  1000  155994    2884  20   0  13492  4136 -      R+   pts/0      0:00  \_ ps
4  1000    1675    1611  20   0 236208  5656 poll_s Ssl+ tty2       0:00 /usr/li
0  1000    1682    1675  20   0 304664 15300 poll_s Sl+  tty2       0:00  \_ /us
```

2. Ascertain how long your system has been up. Display its load averages.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ uptime
 02:31:53 up 22:54,  1 user,  load average: 0.06, 0.02, 0.09
```

3. In your machine environment, how many total running, sleeping, stopped, and zombie processes are there? Type answers below without a captured screen.

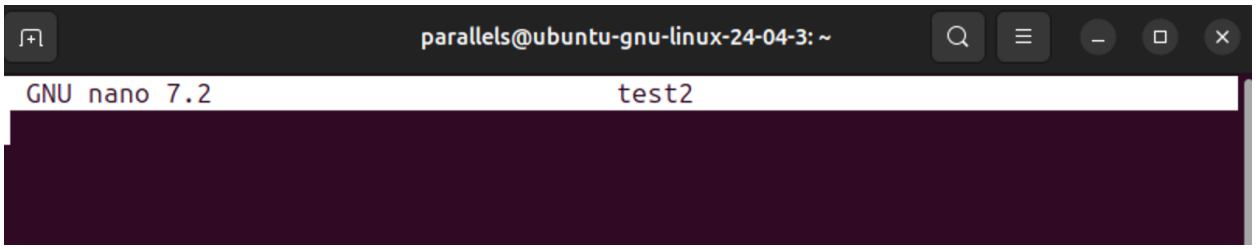
```
top - 02:34:42 up 22:56,  1 user,  load average: 0.00, 0.01, 0.07
Tasks: 217 total,   1 running, 215 sleeping,   0 stopped,   1 zombie
```

4. Create a background process by running command "`nano test1 &`", then list all processes to show that process, then kill that process.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ jobs
[1]-  Stopped                 nano test1
[2]+  Stopped                 top
parallels@ubuntu-gnu-linux-24-04-3:~$ nano test1 &
[2] 156902
parallels@ubuntu-gnu-linux-24-04-3:~$ kill 156902

[2]+  Stopped                 nano test1
```

5. Create a background process by running the command "`nano test2 &`", then bring that process to the foreground process.



6. Run the command "`ps lf`", then change the nice value to be lower priority for the process of the command.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ ps lf
F   UID     PID    PPID PRI  NI    VSZ   RSS WCHAN  STAT TTY        TIME COMMAND
0  1000  157899  157892  20   0  10984  5164 do_wai Ss   pts/0      0:00 bash
0  1000  157942  157899  20   0   8844  2680 do_sig T    pts/0      0:00  \_ nano test2
0  1000  158024  157899  20   0  13492  4136 -      R+   pts/0      0:00  \_ ps lf
4  1000    1675    1611  20   0 236208  5656 poll_s Ssl+ tty2       0:00 /usr/libexec/gdm-wayla
0  1000    1682    1675  20   0 304664 15300 poll_s Sl+  tty2       0:00  \_ /usr/libexec/gnome
parallels@ubuntu-gnu-linux-24-04-3:~$ renice +19 157899
157899 (process ID) old priority 0, new priority 19
parallels@ubuntu-gnu-linux-24-04-3:~$
```

7. Display all processes in real-time using `top` command ordered by the most used memory.

```
top - 02:46:59 up 23:09,  1 user,  load average: 0.02, 0.09, 0.08
Tasks: 213 total,   1 running, 210 sleeping,   1 stopped,   1 zombie
%Cpu(s):  1.7 us,  0.8 sy,  0.0 ni, 97.3 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   3892.4 total,   1159.0 free,   1081.8 used,   1957.6 buff/cache
MiB Swap:   3899.0 total,   3898.7 free,      0.3 used.   2810.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1798 paralle+  20   0 3895512 300788 129940 S   1.3   7.5   3:13.08 gnome-shell
 149345 paralle+  20   0 2813920  62096  46316 S   0.7   1.6   0:17.14 gjs
   2207 paralle+  20   0  920564  58592  44904 S   0.0   1.5   0:00.53 evolution-alarm
 157892 paralle+  20   0  561652  53544  42016 S   0.0   1.3   0:02.52 gnome-terminal-
   5680 root      20   0  633932  44016  37768 S   0.0   1.1   0:03.88 fwupd
   2005 paralle+  20   0 1296196  40600  32284 S   0.0   1.0   0:00.05 evolution-sourc
   2473 paralle+  20   0  921032  40024  29836 S   0.0   1.0   0:01.42 xdg-desktop-por
   2224 paralle+  20   0  180388  38804   5144 S   0.0   1.0   0:01.85 prlcp
 155586 root      20   0 1849896  38316  24472 S   0.0   1.0   0:00.34 snapd
   2522 paralle+  20   0  449912  34464  26884 S   0.0   0.9   0:00.34 snapd-desktop-i
   3009 paralle+  20   0  575416  32360  24784 S   0.0   0.8   0:02.14 update-notifier
   2138 paralle+  20   0  426584  28648  17260 S   0.0   0.7   0:02.37 ibus-extension-
   2345 paralle+  20   0  765764  28564  24524 S   0.0   0.7   0:00.05 evolution-addre
   2201 paralle+  20   0  810252  27084  17936 S   0.7   0.7   0:26.48 prlcc
   2455 paralle+  39  19  811280  26680  17092 S   0.0   0.7   0:00.75 tracker-miner-f
   2454 paralle+  20   0 2587232  26248  21128 S   0.0   0.7   0:00.04 gjs
   2017 paralle+  20   0 2529968  25836  21132 S   0.0   0.6   0:00.02 gjs
```

8. How many total memory, used memory, free memory, and available memory in MB?

```
parallels@ubuntu-gnu-linux-24-04-3:~$ free -mt
               total        used        free      shared  buff/cache   available
Mem:            3892        1090        1149         126        1958        2801
Swap:           3898           0        3898
Total:          7791        1091        5047
```

9. Schedule to run this command in next 5 minutes: `date > /tmp/datestamp`

```
parallels@ubuntu-gnu-linux-24-04-3:~$ at now + 5 minute
warning: commands will be executed using /bin/sh
at Wed Aug 26 02:56:00 2026
at> date > /tmp/datestamp
at> <EOT>
job 3 at Wed Aug 26 02:56:00 2026
```

10. Cronjob
    a. Create a script file named `myjob.sh` containing following by run "nano myjob.sh":

```
#!/bin/bash
echo Hello I am running $0 at $(date) > /tmp/hello-test
```

```
arallels@ubuntu-gnu-linux-24-04-3:~$ nano myjob.sh
```

    b. Make the above file executed by: `chmod +x myjob.sh`

```
ls@ubuntu-gnu-linux-24-04-3:~$ chmod +x myjob.sh
```

    c. Schedule the script to run in the next 5 minutes. Let's note that you may need to check your time on the machine you're using.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ at now + 5 minute
warning: commands will be executed using /bin/sh
at Wed Aug 26 02:58:00 2026
at> myjob.sh
at> <EOT>
job 4 at Wed Aug 26 02:58:00 2026
```

    d. List scheduled jobs to make sure your job has been scheduled.

```
parallels@ubuntu-gnu-linux-24-04-3:~$ atq
4        Wed Aug 26 02:58:00 2026 a parallels
3        Wed Aug 26 02:56:00 2026 a parallels
```